# Who Chooses the Hospital?
# Insurers' Referral Steering and Hospital Prices

Mehrdad Pourghasem

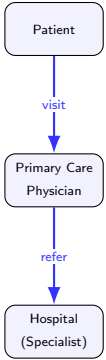
December 8, 2025



Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# Why Referral Steering Matters for Hospital Prices

- Patients rarely choose hospitals

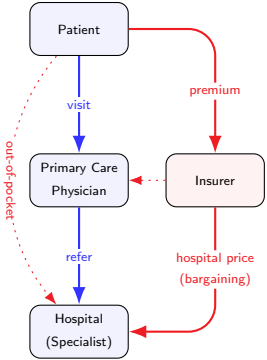





Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# Why Referral Steering Matters for Hospital Prices

- Patients rarely choose hospitals or face hospital prices.

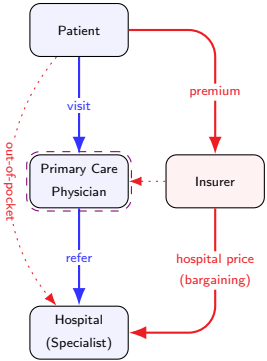





Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# Why Referral Steering Matters for Hospital Prices

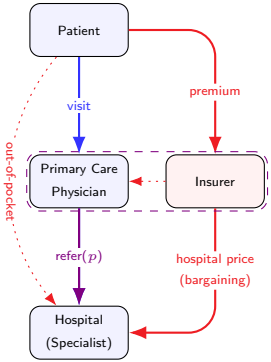
- Patients rarely choose hospitals or face hospital prices.
- **Independent** primary care physicians:
  - Insurers bear hospital prices.
  - Refer based on what is best for the patient.




*Independent* PCPs

# Why Referral Steering Matters for Hospital Prices

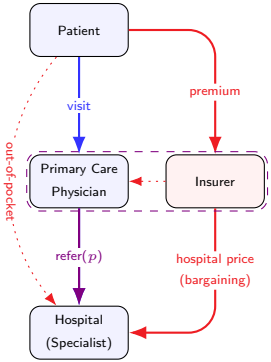
- Patients rarely choose hospitals or face hospital prices.
- **Independent** primary care physicians:
  - Insurers bear hospital prices.
  - Refer based on what is best for the patient.
- **Insurer-aligned** primary care physicians:
  - Refer while taking insurers' interests into account.
  - Referrals become responsive to hospital prices.
  - Hospitals face **more elastic demand**.




*Insurer-aligned* PCPs

# Why Referral Steering Matters for Hospital Prices

- Patients rarely choose hospitals or face hospital prices.
- **Independent** primary care physicians:
  - Insurers bear hospital prices.
  - Refer based on what is best for the patient.
- **Insurer-aligned** primary care physicians:
  - Refer while taking insurers' interests into account.
  - Referrals become responsive to hospital prices.
  - Hospitals face **more elastic demand**.




*Insurer-aligned* PCPs

- **Bargaining implication.** In price bargaining between two parties, higher demand elasticity for a party translates into a weaker bargaining position for that party.
  - In insurer–hospital bargaining: higher hospital demand elasticity ⇒ lower **negotiated prices**.

# Welfare Trade-off from Insurer Referral Steering

- The welfare impact of **insurer referral steering** is theoretically ambiguous.



Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# Welfare Trade-off from Insurer Referral Steering

- The welfare impact of **insurer referral steering** is theoretically ambiguous.
- **Potential Costs:**
  - **Allocative distortions:** patients may be steered to *lower-quality* or *farther-away* hospitals.

# Welfare Trade-off from Insurer Referral Steering

- The welfare impact of **insurer referral steering** is theoretically ambiguous.
- **Potential Costs:**
  - **Allocative distortions:** patients may be steered to *lower-quality* or *farther-away* hospitals.
- **Potential benefits:**
  - **Productive efficiencies:** PCPs internalize treatment costs, leading to *fewer unnecessary procedures*.
  - **Bargaining prices:** lowers hospital prices through bargaining.

# Welfare Trade-off from Insurer Referral Steering

- The welfare impact of **insurer referral steering** is theoretically ambiguous.
- **Potential Costs:**
  - **Allocative distortions:** patients may be steered to *lower-quality* or *farther-away* hospitals.
- **Potential benefits:**
  - **Productive efficiencies:** PCPs internalize treatment costs, leading to *fewer unnecessary procedures*.
  - **Bargaining prices:** lowers hospital prices through bargaining.

**Research Question:**

How does **insurer referral steering** affect **patient flows**, **hospital prices**, and **welfare**?

# Welfare Trade-off from Insurer Referral Steering

- The welfare impact of **insurer referral steering** is theoretically ambiguous.
- **Potential Costs:**
  - **Allocative distortions:** patients may be steered to *lower-quality* or *farther-away* hospitals.
- **Potential benefits:**
  - **Productive efficiencies:** PCPs internalize treatment costs, leading to *fewer unnecessary procedures*.
  - **Bargaining prices:** lowers hospital prices through bargaining.

**Research Question:**

How does **insurer referral steering** affect **patient flows**, **hospital prices**, and **welfare**?

- **Policy counterfactuals:** What happens to prices and welfare if steering is limited?

# How Insurers Steer Referrals and Why

- **How** insurers align PCPs with prices?
  1. **Ownership or integration:** Insurer owns or controls physician groups.



Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# How Insurers Steer Referrals and Why

- **How** insurers align PCPs with prices?
  1. **Ownership or integration:** Insurer owns or controls physician groups.
  2. **Payment contracts:** Capitation (fixed PMPM) paid to PCP groups.

     Under capitation, providers bear financial risk:
     - if total care costs $<$ PMPM, PCPs keep savings.
     - if total care costs $>$ PMPM, PCPs absorb losses.

# How Insurers Steer Referrals and Why

- **How** insurers align PCPs with prices?
  1. **Ownership or integration:** Insurer owns or controls physician groups.
  2. **Payment contracts:** Capitation (fixed PMPM) paid to PCP groups.

     Under capitation, providers bear financial risk:
     - if total care costs $<$ PMPM, PCPs keep savings.
     - if total care costs $>$ PMPM, PCPs absorb losses.
- **Why** not just use networks?
  - Large systems and brand-name hospitals are essential for plan attractiveness.



Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# How Insurers Steer Referrals and Why

- **How** insurers align PCPs with prices?
  1. **Ownership or integration:** Insurer owns or controls physician groups.
  2. **Payment contracts:** Capitation (fixed PMPM) paid to PCP groups.

     Under capitation, providers bear financial risk:
     - if total care costs $<$ PMPM, PCPs keep savings.
     - if total care costs $>$ PMPM, PCPs absorb losses.
- **Why** not just use networks?
  - Large systems and brand-name hospitals are essential for plan attractiveness.
  - Network exclusion is an all-or-nothing decision; dropping an expensive system removes it for all services, not just where it is inefficient.



Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# How Insurers Steer Referrals and Why

- **How** insurers align PCPs with prices?
  1. **Ownership or integration:** Insurer owns or controls physician groups.
  2. **Payment contracts:** Capitation (fixed PMPM) paid to PCP groups.

     Under capitation, providers bear financial risk:
     - if total care costs $<$ PMPM, PCPs keep savings.
     - if total care costs $>$ PMPM, PCPs absorb losses.
- **Why** not just use networks?
  - Large systems and brand-name hospitals are essential for plan attractiveness.
  - Network exclusion is an all-or-nothing decision; dropping an expensive system removes it for all services, not just where it is inefficient.
  - Networks are renegotiated infrequently (multi-year contracts).

# Related Literature

- **Steering via capitation.** Higher capitation shifts referrals toward farther, lower-priced hospitals. (Ho and Pakes 2014)

# Related Literature

- **Steering via capitation.** Higher capitation shifts referrals toward farther, lower-priced hospitals. (Ho and Pakes 2014)
- **Steering via coinsurance.** Higher coinsurance increases patient price sensitivity and lowers hospital prices. (Gowrisankaran, Nevo, and Town 2015)



Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# Related Literature

- **Steering via capitation.** Higher capitation shifts referrals toward farther, lower-priced hospitals. (Ho and Pakes 2014)
- **Steering via coinsurance.** Higher coinsurance increases patient price sensitivity and lowers hospital prices. (Gowrisankaran, Nevo, and Town 2015)
- **What is still missing**
  - The role of **insurer integration with physician groups** as a steering lever beyond capitation.
  - The **benefits** side: efficiency, costs, and premiums—beyond allocative distortions.
  - How **referral price-sensitivity** (when patients do not choose hospitals directly) feeds into **negotiated hospital prices** and **welfare**.

# This Paper

1. Measure **referral price-sensitivity** through **insurer integration with physician groups** and **capitation payment**.

# This Paper

1. Measure **referral price-sensitivity** through **insurer integration with physician groups** and **capitation payment**.
2. Map **referral steering** to **hospital prices** using hospital–insurer Nash-in-Nash bargaining.

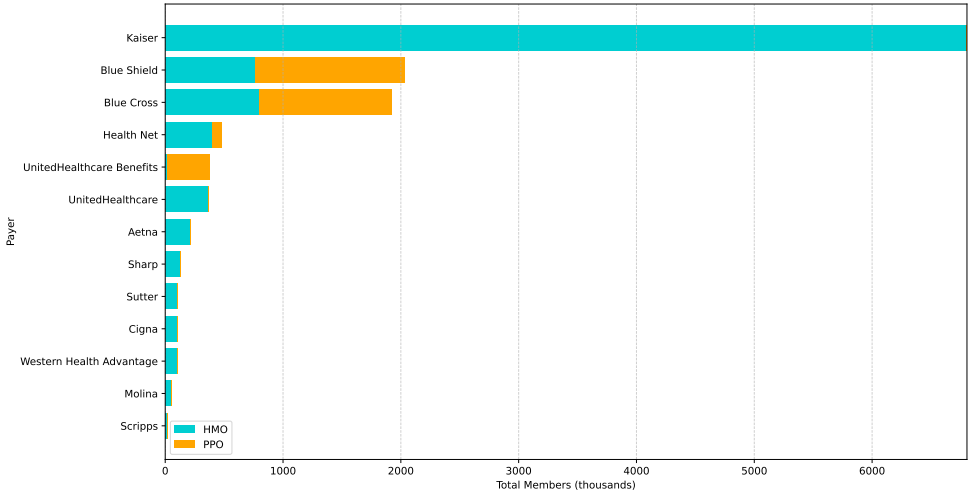
# This Paper

1. Measure **referral price-sensitivity** through **insurer integration with physician groups** and **capitation payment**.
2. Map **referral steering** to **hospital prices** using hospital–insurer Nash-in-Nash bargaining.
3. **Quantify the welfare trade-off** (allocative distortions vs. efficiency/bargaining gains) and evaluates **policy limits** on ownership and capitation.

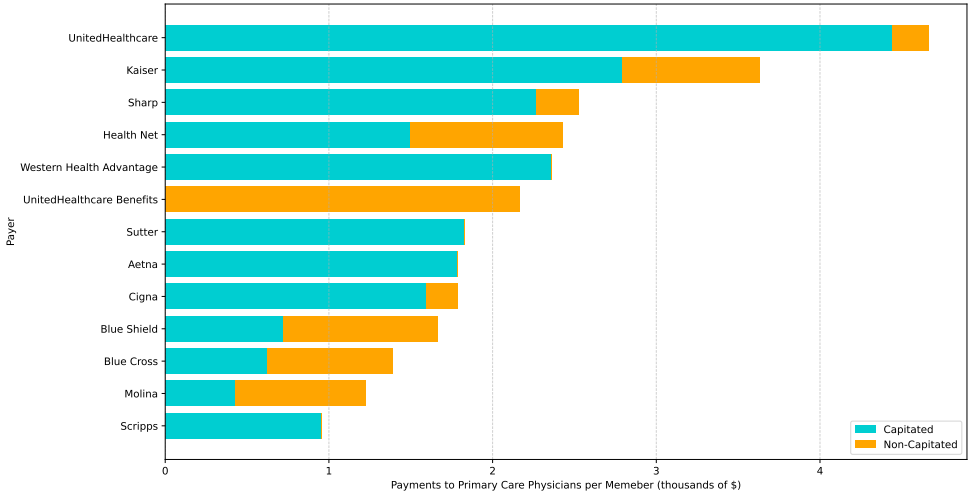
# Key Variables (California 2018–2023)

- **Claims–level data**
  - Patients demographics: age, gender, race, zip.
  - Referral chain: referring provider ID and rendering provider ID.
  - Medical details: date, diagnosis code, procedure code, care setting (inpatient/outpatient).
  - Payer and plan: insurer ID, product type (commercial, medicare advantage), plan name.
  - Payments: total charge, plan payment, copay, deductible, coinsurance, payment arrangement (papitated/fee-for-service).
- **Insurer–level data**
  - Integrated physician groups and their characteristics.
  - Total payments to PCPs, split into capitated vs non-capitated.
- **Hospital–level data**

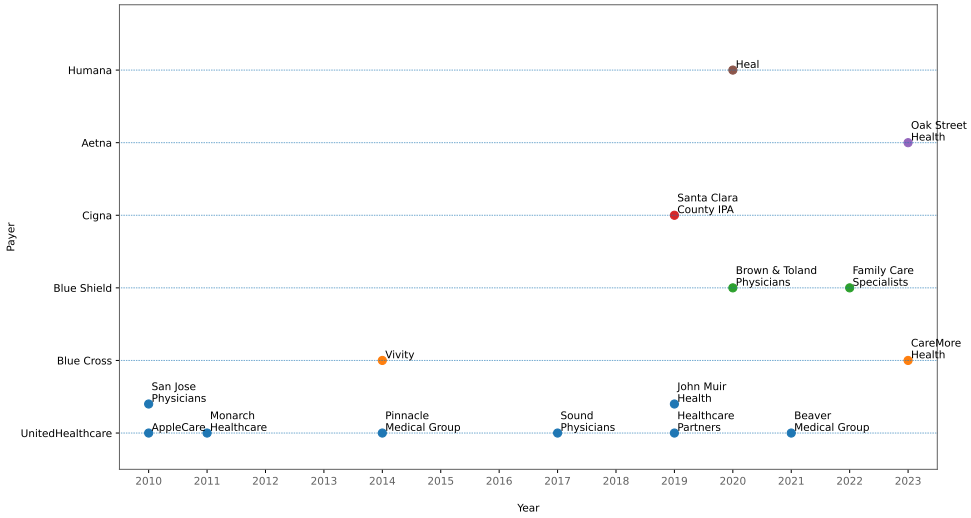
# HMO vs PPO Members



# Capitated vs Non-Capitated Payments to PCPs



# Insurer Ownership and Partnerships with PCP Groups




Dots mark years of acquisitions or partnerships between major insurers and PCP groups.

# Toy Model: Referral-Driven Hospital Demand

- One insurer and two hospitals $h \in \{1, 2\}$.



Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# Toy Model: Referral-Driven Hospital Demand

- One insurer and two hospitals $h \in \{1, 2\}$.
- Patients do not choose hospitals directly: referrals assign a hospital within the network.

# Toy Model: Referral-Driven Hospital Demand

- One insurer and two hospitals $h \in \{1, 2\}$.
- Patients do not choose hospitals directly: referrals assign a hospital within the network.
- Hospital prices affect hospital choice, not patients utility.



Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# Toy Model: Referral-Driven Hospital Demand

- One insurer and two hospitals $h \in \{1, 2\}$.
- Patients do not choose hospitals directly: referrals assign a hospital within the network.
- Hospital prices affect hospital choice, not patients utility.
- **Hospital demand.** Referral-driven quantity to hospital $h$:

$$q_h(\mathcal{N}, \mathbf{p}) = \frac{\exp(v - \theta p_h)}{\sum_{k \in \mathcal{N}} \exp(v - \theta p_k)},$$

  - $v$: patients utility from receiving hospital care.
  - $p_h$: negotiated price with hospital $h$.
  - $\theta$: referrals price sensitivity (how strongly referrals favor cheaper hospitals).
  - $\mathcal{N}$: the set of in-network hospitals.

# Toy Model: Insurer and Hospital Payoffs

- **Insurer objective.** Maximizes enrollees surplus net of payments to hospitals:

$$V(\mathcal{N}, \mathbf{p}) = v + B(\mathcal{N}) - \sum_{h \in \mathcal{N}} p_h \, q_h(\mathcal{N}, \mathbf{p}),$$

  - $B(\mathcal{N})$: value of network breadth.

# Toy Model: Insurer and Hospital Payoffs

- **Insurer objective.** Maximizes enrollees surplus net of payments to hospitals:

$$V(\mathcal{N}, \mathbf{p}) = v + B(\mathcal{N}) - \sum_{h \in \mathcal{N}} p_h \, q_h(\mathcal{N}, \mathbf{p}),$$

  - $B(\mathcal{N})$: value of network breadth.

- **Hospital profits.** Profit of hospital $h$:

$$\pi_h(\mathcal{N}, \mathbf{p}) = (p_h - mc_h) \, q_h(\mathcal{N}, \mathbf{p}),$$

  - $mc_h$: hospital $h$'s marginal cost.

# Toy Model: Nash-in-Nash Bargaining

- Insurer and each hospital $h$ bargain Nash-in-Nash over the price $p_h$:

$$p_h^* = \arg\max_{p_h} \left[\Delta\pi_h\right]^{\tau} \left[\Delta V\right]^{1-\tau},$$

  - $\Delta X \equiv X(\mathcal{N}, \mathbf{p}) - X(\mathcal{N} \setminus \{h\}, p_{-h})$
  - $\tau$: hospital's bargaining weight.

# Toy Model: Nash-in-Nash Bargaining

- Insurer and each hospital $h$ bargain Nash-in-Nash over the price $p_h$:

$$p_h^* = \arg\max_{p_h} \left[\Delta\pi_h\right]^{\tau} \left[\Delta V\right]^{1-\tau},$$

  - $\Delta X \equiv X(\mathcal{N}, \mathbf{p}) - X(\mathcal{N} \setminus \{h\}, p_{-h})$
  - $\tau$: hospital's bargaining weight.

- First-order condition:

$$\tau \frac{\partial \pi_h / \partial p_h}{\Delta \pi_h} = -(1-\tau) \frac{\partial V / \partial p_h}{\Delta V}.$$

where:

$$\frac{\partial \pi_h}{\partial p_h} = q_h - \theta q_h (1 - q_h)(p_h - mc_h),$$
$$\frac{\partial V}{\partial p_h} = -q_h - \theta q_h \left(\overline{p} - p_h\right),$$

  - $\overline{p} \equiv q_1\, p_1 + q_2\, p_2$.

# Toy Model: Bargaining Prices

- Solving the FOC yields:

$$p_h = mc_h + \frac{1}{\underbrace{\theta\,(1-q_h)}_{\substack{\text{direct demand}\\ \text{elasticity}}} + \frac{1-\tau}{\tau}\underbrace{\frac{q_h(1+\theta(\overline{p}-p_h))}{\Delta B + (p_{-h}-\overline{p})}}_{\text{insurer-side elasticity}}}.$$

  - Let $\mu(\theta) \equiv p_h - mc_h = \dfrac{1}{D(\theta)}$. Then, $\dfrac{d\mu}{d\theta} = -\dfrac{D_\theta}{D^2}$.

# Toy Model: Bargaining Prices

- Solving the FOC yields:

$$p_h = mc_h + \frac{1}{\underbrace{\theta\,(1-q_h)}_{\substack{\text{direct demand}\\ \text{elasticity}}} + \frac{1-\tau}{\tau}\underbrace{\frac{q_h(1+\theta(\bar{p}-p_h))}{\Delta B + (p_{-h}-\bar{p})}}_{\text{insurer-side elasticity}}}.$$

  - Let $\mu(\theta) \equiv p_h - mc_h = \dfrac{1}{D(\theta)}$. Then, $\dfrac{d\mu}{d\theta} = -\dfrac{D_\theta}{D^2}$.

**How markup moves with $\theta$**

- Holding prices and shares fixed, the direct effect of $\theta$ on the denominator is:

$$D_\theta = (1-q_h) + \frac{1-\tau}{\tau}\frac{q_h\,(\bar{p}-p_h)}{\Delta B + (p_{-h}-\bar{p})}.$$

  - Near symmetry ($p_h \approx \bar{p}$): the second term is small, so

$$D_\theta \approx 1 - q_h > 0 \Rightarrow \frac{d\mu}{d\theta} < 0,$$

    higher $\theta \to$ lower markup.
  - More generally, as long as the insurer's gain from including hospital $h$, $\Delta B + (p_{-h} - \bar{p})$, is positive and not tiny, the first term dominates, $D_\theta > 0$, so the markup is decreasing in $\theta$.

# Players and Timing

- Players
  - Patients $i \in \mathcal{I}$ with illness $\ell \in \mathcal{L}$.
  - Insurers $m \in \mathcal{M}$ with plans $j \in \mathcal{J}_m$.
  - PCPs $d \in \mathcal{D}$.
  - Hospital systems $s$ with hospitals $h \in \mathcal{H}_s$.

# Players and Timing

- Players
  - Patients $i \in \mathcal{I}$ with illness $\ell \in \mathcal{L}$.
  - Insurers $m \in \mathcal{M}$ with plans $j \in \mathcal{J}_m$.
  - PCPs $d \in \mathcal{D}$.
  - Hospital systems $s$ with hospitals $h \in \mathcal{H}_s$.
- Timing (two stages)

# Players and Timing

- Players
  - Patients $i \in \mathcal{I}$ with illness $\ell \in \mathcal{L}$.
  - Insurers $m \in \mathcal{M}$ with plans $j \in \mathcal{J}_m$.
  - PCPs $d \in \mathcal{D}$.
  - Hospital systems $s$ with hospitals $h \in \mathcal{H}_s$.
- Timing (two stages)
  1. Insurers and hospitals bargain over hospital prices.

# Players and Timing

- Players
  - Patients $i \in \mathcal{I}$ with illness $\ell \in \mathcal{L}$.
  - Insurers $m \in \mathcal{M}$ with plans $j \in \mathcal{J}_m$.
  - PCPs $d \in \mathcal{D}$.
  - Hospital systems $s$ with hospitals $h \in \mathcal{H}_s$.
- Timing (two stages)
  1. Insurers and hospitals bargain over hospital prices.
  2. Patients realize health shocks and they either

# Players and Timing

- Players
  - Patients $i \in \mathcal{I}$ with illness $\ell \in \mathcal{L}$.
  - Insurers $m \in \mathcal{M}$ with plans $j \in \mathcal{J}_m$.
  - PCPs $d \in \mathcal{D}$.
  - Hospital systems $s$ with hospitals $h \in \mathcal{H}_s$.
- Timing (two stages)
  1. Insurers and hospitals bargain over hospital prices.
  2. Patients realize health shocks and they either
     2.1 directly choose a hospital within their plan network, or

# Players and Timing

- Players
  - Patients $i \in \mathcal{I}$ with illness $\ell \in \mathcal{L}$.
  - Insurers $m \in \mathcal{M}$ with plans $j \in \mathcal{J}_m$.
  - PCPs $d \in \mathcal{D}$.
  - Hospital systems $s$ with hospitals $h \in \mathcal{H}_s$.
- Timing (two stages)
  1. Insurers and hospitals bargain over hospital prices.
  2. Patients realize health shocks and they either
     - 2.1 directly choose a hospital within their plan network, or
     - 2.2 see a PCP within network; the PCP (with patient) selects among affiliated in-network hospitals.

# Model Overview

1. Patient direct hospital choice
   - Hospital demand elasticity of negotiated price through coinsurance.

# Model Overview

1. Patient direct hospital choice
   - Hospital demand elasticity of negotiated price through coinsurance.
2. Referral choice
   - Hospital demand elasticity of negotiated price
     - 2.1 for patients enrolled in high-capitation vs. low-capitation insurers.
     - 2.2 for PCPs integrated with insurers vs. non-integrated with insurers.

# Model Overview

1. Patient direct hospital choice
   - Hospital demand elasticity of negotiated price through coinsurance.
2. Referral choice
   - Hospital demand elasticity of negotiated price
     - 2.1 for patients enrolled in high-capitation vs. low-capitation insurers.
     - 2.2 for PCPs integrated with insurers vs. non-integrated with insurers.
3. Hospital–insurer price negotiation
   - Map insurer referral steering to negotiated price.

# Model Overview

- Patient $i$ belongs to plan $j$, which is fixed, having been chosen by long-term employer–insurer contracts.

# Model Overview

- Patient $i$ belongs to plan $j$, which is fixed, having been chosen by long-term employer–insurer contracts.
- Given plan enrollment and network composition, patient $i$ experiences illness $\ell$ with probability $f_{i\ell}$.

# Model Overview

- Patient $i$ belongs to plan $j$, which is fixed, having been chosen by long-term employer–insurer contracts.
- Given plan enrollment and network composition, patient $i$ experiences illness $\ell$ with probability $f_{i\ell}$.
- Insurer $m$'s in-network hospitals: $\mathcal{N}_m \subseteq \mathcal{H}$.

# Model Overview

- Patient $i$ belongs to plan $j$, which is fixed, having been chosen by long-term employer–insurer contracts.
- Given plan enrollment and network composition, patient $i$ experiences illness $\ell$ with probability $f_{i\ell}$.
- Insurer $m$'s in-network hospitals: $\mathcal{N}_m \subseteq \mathcal{H}$.
- For each $m$ and $h \in \mathcal{N}_m$, there is a negotiated **base price** $p_{mh}$.

# Model Overview

- Patient $i$ belongs to plan $j$, which is fixed, having been chosen by long-term employer–insurer contracts.
- Given plan enrollment and network composition, patient $i$ experiences illness $\ell$ with probability $f_{i\ell}$.
- Insurer $m$'s in-network hospitals: $\mathcal{N}_m \subseteq \mathcal{H}$.
- For each $m$ and $h \in \mathcal{N}_m$, there is a negotiated **base price** $p_{mh}$.
- The price paid by insurer $m$ for treating condition $\ell$ at hospital $h$ is $w_\ell \, p_{mh}$, where $w_\ell$ is the resource-intensity weight.

# Patient Direct Hospital Choice

- For each realized illness $\ell$, the patient seeks hospital care directly within her plan network.



Introduction
Data
Toy Model
Model
Model Overview
Hospital Demand
Profit Equations
Price Negotiation
Estimation

# Patient Direct Hospital Choice

- For each realized illness $\ell$, the patient seeks hospital care directly within her plan network.
- The utility that patient $i$ receives from care at hospital $h \in \mathcal{N}_m$:

$$u_{i(\text{direct})h\ell} = \underbrace{\boldsymbol{\beta}^{\text{direct}} \mathbf{x}_{ih\ell} - \alpha_{i\ell}^{\text{direct}} c_{i\ell} w_\ell p_{m(i)h}}_{\delta_{ih\ell}} + \varepsilon_{ih\ell},$$

  - $\mathbf{x}_{ih\ell}$: vector of hospital and patient characteristics (e.g. travel time, hospital-year FEs and interactions of hospital FEs with $w_\ell$).
  - $c_{i\ell}$: coinsurance rate for patient $i$ with illness $\ell$.
  - $\varepsilon_{ih\ell}$: i.i.d. error term that is distributed type 1 extreme value.

# Referral Choice

- For each realized illness $\ell$, the patient sees a PCP $d(i)$. The PCP refers the patient to the hospital which maximize their joint utility.

# Referral Choice

- For each realized illness $\ell$, the patient sees a PCP $d(i)$. The PCP refers the patient to the hospital which maximize their joint utility.
- The utility that patient $i$ and PCP $d(i)$ receive from care at hospital $h \in \mathcal{N}_m$:

$$u_{i(\text{referral})h\ell} = \underbrace{\boldsymbol{\beta}^{\text{referral}}\,\mathbf{x}_{ih\ell} - \alpha_{i\ell}^{\text{referral}}\,c_{i\ell}\,w_\ell\,p_{m(i)h}}_{v^i_{ih\ell}\text{: patient part}}$$

$$\underbrace{-\eta_{i\ell}\,b_{m(i)}\,w_\ell\,p_{m(i)h} - \theta_{i\ell}\,\mathbf{1}\{\mathrm{I}_{m(i)d(i)}\}\,w_\ell\,p_{m(i)h}}_{v^d_{ih\ell}\text{: steering part}} + \varepsilon_{ih\ell},$$

  - $b_{m(i)}$: insurer's $m$ PCPs capitation payment intensity.
  - $\mathbf{1}\{\mathrm{I}_{m(i)d(i)}\} = 1$ if PCP $d$ is integrated with insurer $m$.

# Expected Utility and Hospital Demand

- The hospital **choice probability** for patient $i$ with illness $\ell$:

$$s_{i(\text{direct})h\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \frac{\exp\left(\delta_{ih\ell}\right)}{\sum_{k\in\{0\}\cup\mathcal{N}_{m(i)}} \exp\left(\delta_{ik\ell}\right)},$$

$$s_{i(\text{referral})h\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \frac{\exp\left(v^i_{ih\ell} + v^d_{ih\ell}\right)}{\sum_{k\in\{0\}\cup\mathcal{N}_{m(i)}} \exp\left(v^i_{ik\ell} + v^d_{ik\ell}\right)}.$$

# Expected Utility and Hospital Demand

- The hospital **choice probability** for patient $i$ with illness $\ell$:

$$s_{i(\text{direct})h\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \frac{\exp\left(\delta_{ih\ell}\right)}{\sum_{k\in\{0\}\cup\mathcal{N}_{m(i)}} \exp\left(\delta_{ik\ell}\right)},$$

$$s_{i(\text{referral})h\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \frac{\exp\left(v^i_{ih\ell} + v^d_{ih\ell}\right)}{\sum_{k\in\{0\}\cup\mathcal{N}_{m(i)}} \exp\left(v^i_{ik\ell} + v^d_{ik\ell}\right)}.$$

- The ex ante **expected utility** to patient $i$:

$$W_{i(\text{direct})}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \sum_{\ell} f_{i\ell} \sum_{h\in\{0\}\cup\mathcal{N}_{m(i)}} s_{ih\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) \delta_{ih\ell},$$

$$W_{i(\text{referral})}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \sum_{\ell} f_{i\ell} \sum_{h\in\{0\}\cup\mathcal{N}_{m(i)}} s_{ih\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) v^i_{ih\ell}.$$

# Expected Utility and Hospital Demand

- The hospital **choice probability** for patient $i$ with illness $\ell$:

$$s_{i(\text{direct})h\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \frac{\exp\left(\delta_{ih\ell}\right)}{\sum_{k\in\{0\}\cup\mathcal{N}_{m(i)}} \exp\left(\delta_{ik\ell}\right)},$$

$$s_{i(\text{referral})h\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \frac{\exp\left(v^i_{ih\ell} + v^d_{ih\ell}\right)}{\sum_{k\in\{0\}\cup\mathcal{N}_{m(i)}} \exp\left(v^i_{ik\ell} + v^d_{ik\ell}\right)}.$$

- The ex ante **expected utility** to patient $i$:

$$W_{i(\text{direct})}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \sum_\ell f_{i\ell} \sum_{h\in\{0\}\cup\mathcal{N}_{m(i)}} s_{ih\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) \delta_{ih\ell},$$

$$W_{i(\text{referral})}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) = \sum_\ell f_{i\ell} \sum_{h\in\{0\}\cup\mathcal{N}_{m(i)}} s_{ih\ell}\left(\mathcal{N}_{m(i)}, \mathbf{p}_{m(i)}\right) v^i_{ih\ell}.$$

- The normalized **quantity** to hospital $h$ from insurer $m$ (scales with severity $w_\ell$):

$$q_{mh}\left(\mathcal{N}_m, \mathbf{p}_m\right) = \sum_i \sum_\ell \mathbf{1}\{m(i) = m\}\, f_{i\ell}\, w_\ell\; s_{ih\ell}\left(\mathcal{N}_m, \mathbf{p}_m\right).$$

# Insurer Objective (acting for employers)

- Each insurer, on behalf of its contracted employers, maximizes a weighted sum of enrollees' consumer surplus, net of payments to hospitals, holding $m(i)$ fixed.

# Insurer Objective (acting for employers)

- Each insurer, on behalf of its contracted employers, maximizes a weighted sum of enrollees' consumer surplus, net of payments to hospitals, holding $m(i)$ fixed.
- The ex ante expected cost to the insurer and its employer:

$$TC_m\left(\mathcal{N}_m, \mathbf{p}_m\right) = \sum_i \sum_\ell \mathbf{1}\{m(i) = m\}(1 - c_{i\ell})\, f_{i\ell}\, w_\ell \sum_{h \in \{0\} \cup \mathcal{N}_m} p_{mh}\, s_{ih\ell}\left(\mathcal{N}_m, \mathbf{p}_m\right),$$

# Insurer Objective (acting for employers)

- Each insurer, on behalf of its contracted employers, maximizes a weighted sum of enrollees' consumer surplus, net of payments to hospitals, holding $m(i)$ fixed.
- The ex ante expected cost to the insurer and its employer:

$$TC_m\left(\mathcal{N}_m, \mathbf{p}_m\right) \;=\; \sum_i \sum_\ell \mathbf{1}\{m(i)=m\}(1-c_{i\ell})\, f_{i\ell}\, w_\ell \sum_{h\in\{0\}\cup\mathcal{N}_m} p_{mh}\, s_{ih\ell}\left(\mathcal{N}_m, \mathbf{p}_m\right),$$

- The value (in dollars) to the insurer and the employer it represents:

$$V_m\left(\mathcal{N}_m, \mathbf{p}_m\right) \;=\; \frac{\gamma}{\alpha} \sum_i \mathbf{1}\{m(i)=m\}\, W_i\left(\mathcal{N}_m, \mathbf{p}_m\right) \;-\; TC_m\left(\mathcal{N}_m, \mathbf{p}_m\right).$$

  - $\gamma$: relative weight on employee welfare.

# Hospital Profit

- Hospital's profit:

$$\pi_s\left(\mathcal{M}_s, \{\mathbf{p}_m\}_{m\in\mathcal{M}_s}, \{\mathcal{N}_m\}_{m\in\mathcal{M}_s}\right) = \sum_{m\in\mathcal{M}_s}\sum_{h\in\mathcal{H}_s} q_{mj}\left(\mathcal{N}_m, \mathbf{p}_m\right)\left[p_{mh} - mc_{mh}\right],$$

  - $\mathcal{M}_s$: set of insurers that include system $s$ in their network.
  - $mc_{mh}$: constant marginal cost scaled by possible insurer-specific cost shifters.

# Price Negotiation

- Nash-in-Nash bargaining problem for insurer $m$ and hospital system $s$:

$$p^*_{mh} = \arg\max_{p_{mh}} \left[ \sum_{h \in \mathcal{H}_s} q_{mh}\left(\mathcal{N}_m, \mathbf{p}_m\right)\left[p_{mh} - mc_{mh}\right] \right]^{\tau_{s(m)}} \left[ V_m\left(\mathcal{N}_m, \mathbf{p}_m\right) - V_m\left(\mathcal{N}_m \backslash \mathcal{H}_s, \mathbf{p}_m \backslash p_{mh}\right) \right]^{\tau_{m(s)}},$$

  - Bargaining weights: $\tau_s(m)$ for the system, $\tau_m(s)$ for the insurer (sum to 1).
  - For each $(m, s)$ contract, solve a Nash bargaining problem over $\{p_{mh}\}_{h \in \mathcal{H}_s}$, conditioning on all other contracts.
  - Disagreement values reflect removal of the system (or insurer) from the network.

# First-Order Conditions and Price Formula

- Solving FOCs yields:

$$\mathbf{p} = \mathbf{mc} - (\mathbf{\Omega} + \mathbf{\Lambda})^{-1}\mathbf{q},$$

  - $\Omega_{jk} = \partial q_{mk}/\partial p_{mj}$: actual price sensitivity.
  - $\Lambda$ captures how the insurer's value responds to price changes, internalizing patient steering.

- Interpretation: identical to inverse-elasticity markup rule, but with *effective* sensitivity $(\Omega + \Lambda)$.
- Effective own-price sensitivity $>$ actual patient sensitivity when MCOs can steer, implying lower markups than pure Bertrand.

# Estimation Strategy

- **Stage 2 (demand):** Multinomial logit using claims data; price enters via coinsurance and referral price-sensitivity.
- **Stage 1 (bargaining):** GMM on inverted FOCs to recover $mc$, $\tau$, and bargaining parameters.

# Appendix

# References I

Gowrisankaran, Gautam, Aviv Nevo, and Robert Town (2015). "Mergers when prices are negotiated: Evidence from the hospital industry". In: *American Economic Review* 105.1, pp. 172–203.

Ho, Kate and Ariel Pakes (2014). "Hospital choices, hospital prices, and financial incentives to physicians". In: *American Economic Review* 104.12, pp. 3841–3884.